# ActiveVideoAgent: Agent-Driven Spatio-Temporal Indexing for Long Video Reasoning

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***Abstract*—Video question answering (Video QA) requires effective reasoning over long-duration, multimodal content. While retrieval-augmented generation (RAG) shows promise, existing methods struggle with long-range temporal dependencies and multi-step inference. This paper proposes ActiveVideoAgent (AVA), an agent-driven multimodal retrieval-generation framework, which combines a structured multimodal index (STC-Index) with agent-driven dynamic retrieval and step-wise prompting to enable selective evidence aggregation while preserving temporal coherence. On the LongVideoReasoning benchmark, AVA outperforms representative baselines on 4 of 5 evaluation dimensions. Ablation studies further validate the contributions of the agent-driven retrieval mechanism and the structured indexing design to multimodal evidence integration.**

***Keywords*—video question answering, retrieval-augmented generation, long video understanding, multi-step reasoning**

## I. Introduction

Video question answering (Video QA) requires reasoning over temporally extended multimodal evidence to answer queries [1]. Recent retrieval-augmented generation (RAG) methods have improved video-text alignment and evidence grounding for short-clip QA [2], [3]. However, long-video QA remains fundamentally challenging [4]. Existing systems treat video content as independent clips and retrieve evidence through static similarity matching, which disrupts long-range temporal dependencies, fragments context across distant segments, exacerbates cross-modal conflicts, and fails to support the multi-step logical inference required by complex queries [1], [5], [6], [7].

To address the above challenges, we propose **ActiveVideoAgent** (AVA), an agent-driven multimodal retrieval-generation framework for long-video QA. AVA shifts from flat retrieval to a structured multimodal indexing system (STC-Index). By extracting features via Whisper [8], MiniCPM-V [9], and ImageBind [10], AVA organizes visual vectors, entity-relation graphs, and hierarchical text into a cohesive temporal structure. Coupled with dynamic multi-channel retrieval [11] and structured Chain-of-Thought (CoT) prompting [12], [7], AVA explicitly grounds Large Language Models (LLMs) to enable continuous, cross-modal logical inference over extended video contexts. The main contributions of this work are summarized as follows:

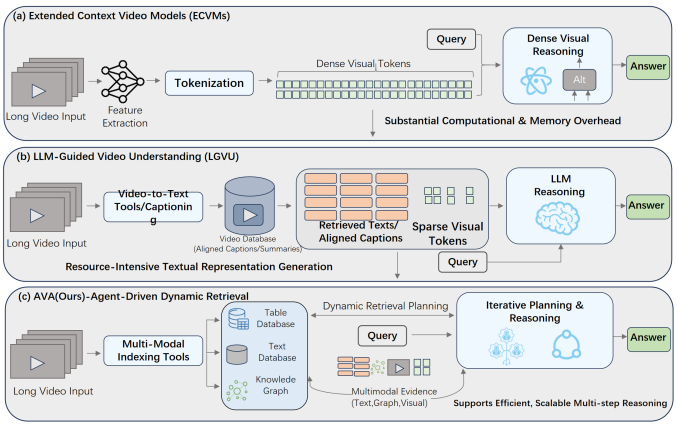



Fig. 1. Comparison of long-video QA paradigms.

- **STC-Index:** A structured multimodal index that organizes visual, textual, and semantic information to preserve temporal dependencies and resolve cross-modal conflicts.
- **Agent-Driven Retrieval:** A dynamic routing mechanism that adaptively selects retrieval channels and refines queries to resolve multi-hop temporal and causal dependencies.
- **Hierarchical CoT Prompting:** A structured generation strategy that leverages retrieved evidence to guide complex, multi-step logical inference.

## II. Related Work

Recent advancements in RAG mitigate LLM hallucinations by dynamically querying external knowledge [13]. While text-based RAG incorporates planning and reflection [14], Multimodal RAG (M-RAG) extends this framework to heterogeneous streams via cross-modal alignment and graph-based indexing [4], [15]. However, applying M-RAG to long-video QA poses severe challenges due to context limits and exponential computational costs [16], [17].

Existing long-video paradigms involve critical structural trade-offs. As shown in Figure 1(a), Extended Context Vision Models (ECVMs) [18], [19], [20], [21] which directly process dense visual tokens, preserve temporal dynamics but incur prohibitive costs and struggle to resolve explicit cross-modal

TABLE I. STRUCTURAL COMPARISON AGAINST REPRESENTATIVE RAG PARADIGMS.

| Paradigm | Evidence Space | Controller Policy | Retrieval Loops | Temporal Constraint |
|---|---|---|---|---|
| GraphRAG | Textual Entity Graph | Static Similarity | Single-shot | None |
| VideoRAG | Video Segments | Static Multi-channel | Single-shot | Weak |
| ReAct-based | Free Text Tools | Unconstrained Reflection | Multi-turn | None |
| Ours | Multimodal Graph + Video | Explicit Relevance Router | Adaptive w/ Threshold | Strict Anchoring |

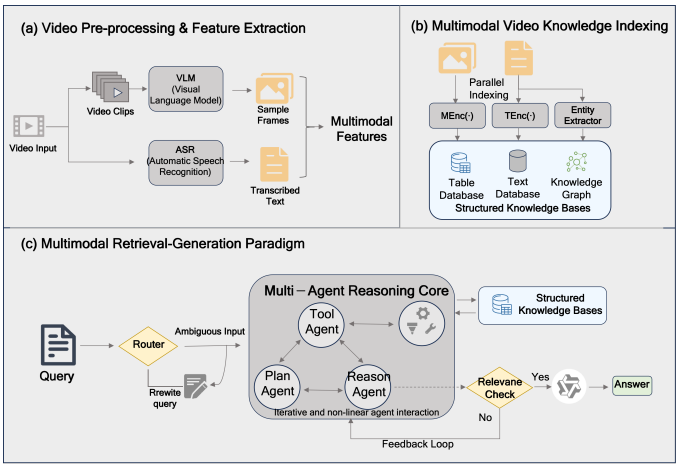



Fig. 2. Overview of the proposed AVA framework.

conflicts [22], [11]. Conversely, LLM-Guided Video Understanding (LGVU) pipelines, as shown in Figure 1(b), compress multimodal streams into discrete text, severely degrading inherent temporal continuity [23], [24], [25]. Furthermore, standard VideoRAG predominantly relies on static similarity matching, which disrupts the long-range temporal dependencies crucial for complex multi-hop inference [26], [27], [28]. To bridge these gaps, AVA replaces flat text conversion and heuristic matching with structured multimodal indexing and agent-driven dynamic retrieval.

## III. METHODOLOGY

We formulate long-video QA as a structured M-RAG problem. Given a video $\mathcal{V}$ and query $q$, the objective is to generate an answer $\mathcal{A}$ preserving temporal consistency, cross-modal alignment, and logical coherence: $\mathcal{I} = \textsc{Index}(\mathcal{V})$, $\mathcal{E}^\star = \textsc{Retrieve}_\pi(q, \mathcal{I})$, and $\mathcal{A} = \textsc{Generate}(q, \mathcal{E}^\star)$, where $\mathcal{I}$ is the multimodal index (STC-index), $\pi$ represents an agent policy, and $\mathcal{E}$ is the evidence set. Figure 2 and Table I illustrate our overall paradigm compared to existing approaches.

After video preprocessing and feature extraction via automatic speech recognition (ASR), a VLM model for visual captions, and an embedding model for cross-modal embeddings, the STC-index partitions video $\mathcal{V}$ into temporally ordered records $\mathcal{S} = \{s_j\}_{j=1}^{m}$, where each $s_j = (t_j, F_j, x_j, c_j)$ contains the timestamp interval $t_j$, sampled frames $F_j$, ASR transcript $x_j$, and visual caption $c_j$. These populate three coordinated indices $\mathcal{I} = \{I_v, I_t, \mathcal{G}\}$.

**Visual index** $I_v = \{(j, E_j^v, t_j)\}$ stores time-stamped segment embeddings, where $E_j^v = \phi_v(F_j)$ denotes the visual embedding computed from the sampled frames. **Text index** serializes $x_j$ and $c_j$, yielding $I_t = \{(j, x_j \oplus c_j, t_j)\}$. **Graph index** captures structured semantics, with entities $\mathcal{N}_j$ and relations $\mathcal{R}_j$ extracted via schema-constrained LLMs to ensure structural consistency across segments, yielding $\mathcal{G} = \{\mathcal{G}_j\} = \{(\mathcal{N}_j, \mathcal{R}_j)\}$. Cross-modal conflict is flagged when $\textsc{Conflict}(s_j) = \mathbf{1}\big[\cos\big(E_j^v, E_j^t\big) < \delta\big]$, where $E_j^t = \phi_t(x_j \oplus c_j)$ denotes the text embedding.

Retrieval is driven by an agent acting as a sequential decision process. At step $t$, the state $z_t = (q, H_t, \mathcal{E}_t)$ includes the query, retrieval history $H_t$, and accumulated evidence $\mathcal{E}_t$. An action $a_t = (T_t, \hat{q}_t)$ selects a tool $T_t \in \{I_v, I_t, \mathcal{G}\}$ and optionally rewrites the query. For a rewritten query $\hat{q}_t$, candidate evidence $\mathcal{E}'$ is scored via:

$$\text{Score}(\mathcal{E}') = \lambda_1 S_{\text{sem}} + \lambda_2 S_{\text{ctx}} + \lambda_3 S_{\text{cred}} \tag{1}$$

where $S_{\text{sem}}$ measures embedding similarity, $S_{\text{ctx}}$ enforces temporal consistency and cross-modal agreement, and $S_{\text{cred}}$ measures graph reliability. The agent explicitly maximizes feedback by selecting $T_t = \arg\max_T \text{Score}(\mathcal{E}_t)$, terminating when $\text{Score}(\mathcal{E}_t) \geq \tau$. Final evidence aggregation merges these streams via $\mathcal{E}^\star = \bigcup_c w_c \cdot \mathcal{E}_c$, where channel weights $w = \text{softmax}(W\phi_t(q))$ are query-conditioned.

During generation, each retained segment $\hat{s}_j \in \mathcal{E}^\star$ is serialized into an evidence block $\hat{C}_j = \textsc{Fuse}(t_j, x_j, c_j, F_j, \mathcal{G}_j)$. Blocks are sorted chronologically, and the final answer is generated as $\hat{y} = \text{LLM}(\textsc{Prompt}(q, \text{Seq}(\{\hat{C}_j\})))$. To enforce temporal and multimodal constraints, the prompt strictly dictates answering based on evidence blocks anchored to timestamps (e.g., "[Time: $t_i$-$t_j$]") and structured relational triples, instructing the model to perform step-by-step chronological reasoning with explicit cross-modal conflict resolution.

## IV. EXPERIMENTS

We evaluate our framework on LongVideoReasoning [3], a large-scale benchmark specifically designed for long-range temporal and multi-hop video reasoning. It comprises 2,184 QA pairs (1,520 train / 664 test) spanning videos with an average duration of 45.6 minutes. We extract 5 keyframes per segment for initial indexing, dynamically expanding to 15 frames via scene-change detection during the retrieval phase. ImageBind [10] is utilized for embedding, Whisper [8] for ASR, MiniCPM-V [9] for visual captioning, and OpenAI's `text-embedding-3-small` for semantic retrieval. The core indexing, agent routing, and answer generation are uniformly driven by Qwen2.5-72B-Instruct.

Mapping to our formulated policy, the relevance score weights $\lambda_1$, $\lambda_2$ and $\lambda_3$ are empirically set to 0.5, 0.3, and 0.2, respectively. These weights, alongside the threshold $\tau = 0.65$, were determined via grid-search on 150 validation samples. Routing retrieves $k_v = 4$ video segments and $k_t = 2$ text chunks per step. All experiments are executed on 8× NVIDIA A100 (80GB) GPUs using FP16 precision.

We compare AVA with existing RAG architectures: NaiveRAG, GraphRAG, LightRAG and VideoRAG. For strict fairness, all baselines are re-implemented using identical foundational models and data processing. Due to the limits of n-gram metrics in logical reasoning, we adopt a standardized LLM-as-a-Judge protocol using Qwen-2.5-72B. The reliability of the automated judge is validated by 100 human annotations (Cohen's $\kappa = 0.78$), confirming its alignment with human judgments. We employ two evaluation formats: absolute scoring on a 1-5 scale across five dimensions and pairwise win-rate with alternating order to mitigate positional bias.

TABLE II. OVERALL ABSOLUTE PERFORMANCE COMPARISON ACROSS ALL VIDEO CATEGORIES (1-5 SCALE, HIGHER IS BETTER).

| Metric | GraphRAG | LightRAG | VideoRAG | AVA |
|---|---|---|---|---|
| Comprehensiveness | 3.64 | 3.65 | 4.04 | **4.09** |
| Empowerment | 3.72 | 3.71 | 4.07 | **4.13** |
| Trustworthiness | 3.66 | 3.70 | 4.04 | **4.08** |
| Depth | 3.60 | 3.61 | 4.09 | **4.21** |
| Density | **4.20** | 4.19 | 4.16 | 4.19 |
| Overall Score | 3.67 | 3.67 | 4.04 | **4.09** |

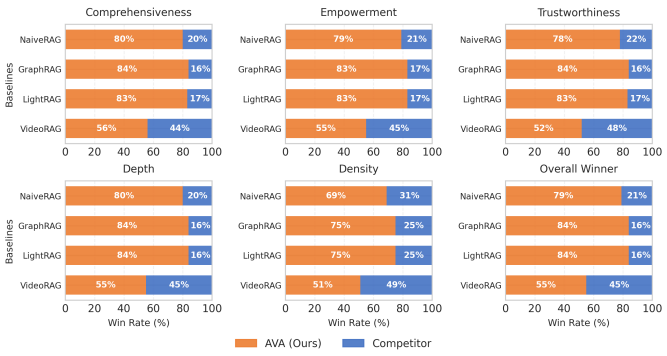



Fig. 3. Win-rate comparison of AVA and baseline RAG methods.

## A. Comparison with Baselines

Table II presents the absolute scores across different dimensions. AVA achieves the highest overall score of 4.09, consistently outperforming the strongest baseline, VideoRAG (4.04). The performance margins are particularly pronounced in the Depth (4.21) and Empowerment (4.13) dimensions. This demonstrates that our explicitly anchored temporal indexing and multi-step agent routing effectively mitigate the contextual fragmentation often observed in standard dense retrieval. Furthermore, as illustrated in Figure 3, AVA maintains a dominant win-rate against all baseline configurations across Lecture, Documentary, and Entertainment domains. The pronounced advantage in Documentaries, which inherently contain complex, non-linear narratives, further validates the robustness of our dynamic multimodal graph traversal.

## B. Temporal Reasoning Validation

To verify that performance gains arise from deep temporal reasoning rather than shallow semantic matching, we analyze pairwise win-rates against VideoRAG across temporal span (time gap between the earliest and latest evidence) and hop count (number of required evidence segments). As Figure 4 shows, AVA's quantitative advantage increases with task complexity. Figure 4(a) indicates comparable performance on localized queries (span $\leq$ 2 minutes) with a 54.4% win-rate. However, on long-range spans ($>$ 10 minutes), the win-rate jumps significantly to 60.5%. Figure 4(b) shows a 54.3% win-rate for shallow 1-2 hop queries, which surges to 67.6% for highly complex reasoning (5+ hops). This superiority in high-hop scenarios indicates that our agent-driven iterative retrieval accurately tracks state changes across multiple independent segments. Figure 4(c) reveals the largest advantage (69% win-rate) occurs in the most challenging zone: large spans with high hops ($>$ 10m, 5+ hops).

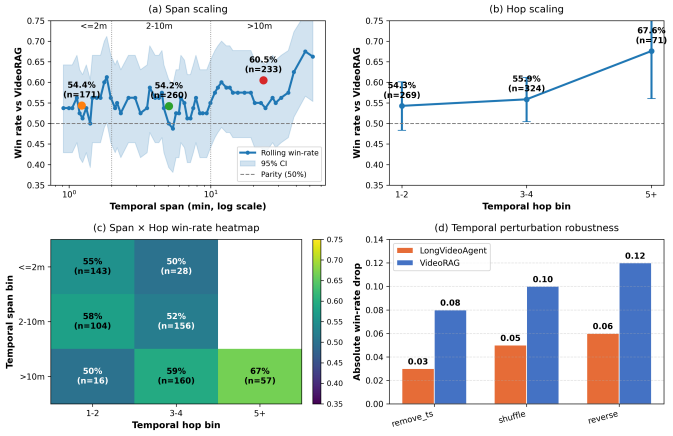



Fig. 4. Temporal reasoning diagnosis (win-rate vs. VideoRAG). (a) Span scaling: rolling win-rate across temporal spans. (b) Hop scaling: win-rate across multi-hop bins. (c) Span×Hop: win-rate heatmap in complex intersection zones (brackets denote sample size). (d) Temporal perturbation: absolute performance drop under missing timestamps, shuffled, and reversed evidence.

Additionally, we evaluate robustness against temporal structural perturbations. Given an initially chronologically ordered evidence sequence $\mathcal{E} = \{s_1, s_2, \ldots, s_m\}$ and its timestamp set $\mathcal{T} = \{t_1, t_2, \ldots, t_m\}$, we design three input-level perturbations:

- Remove timestamps (remove_ts): Mask explicit timestamp metadata in text inputs, i.e., $s'_i = s_i \setminus \{t_i\}$.
- Shuffle order (shuffle): Apply a random permutation $\pi$ to the evidence sequence, yielding $\mathcal{E}_{\text{shuffle}} = \{s_{\pi(1)}, s_{\pi(2)}, \ldots, s_{\pi(m)}\}$.
- Reverse input (reverse): Invert the input sequence order completely, i.e., $\mathcal{E}_{\text{reverse}} = \{s_m, s_{m-1}, \ldots, s_1\}$.

As shown in Figure 4(d), VideoRAG suffers catastrophic performance drops under these extreme noise conditions (falling by 0.10 and 0.12 for shuffle and reverse, respectively), exposing its over-reliance on absolute text input order. AVA's drops are limited to 0.05 and 0.06. By explicitly anchoring temporal and causal relations within the structured STC-Index, the model achieves permutation invariance. It dynamically reconstructs the correct event topology to complete reasoning, even when the input sequence is completely disrupted.

## C. Cross-modal Conflict Resolution

To evaluate evidence integration and conflict resolution under visual-textual misalignment, we map each video segment $s_j = (t_j, F_j, x_j, c_j)$ into a shared cross-modal embedding space. This yields visual representations $E_j^v = \phi_v(F_j)$ and textual representations $E_j^t = \phi_t(x_j \oplus c_j)$. The segment-level conflict intensity is defined as:

$$\tilde{\text{conf}}_j = \text{clip}\left(\frac{1 - \cos(E_j^v, E_j^t)}{2}, 0, 1\right), \tag{2}$$

where higher values indicate weaker cross-modal consistency. For a query $q$ with the final evidence set $\mathcal{E}(q) = \{s_{j_1}, \ldots, s_{j_m}\}$, we normalize the retrieval scores $\text{Score}_{j_i}$ into weights:

$$\alpha_i = \frac{\exp(\text{Score}_{j_i})}{\sum_{k=1}^{m} \exp(\text{Score}_{j_k})}, \tag{3}$$

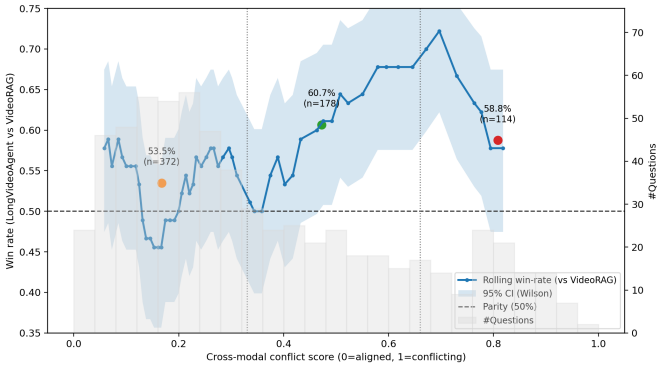



Fig. 5. Performance evaluation under varying cross-modal conflict intensities. The rolling win-rate demonstrates AVA's superior stability on highly contradictory samples compared to VideoRAG.

and define the query-level conflict score as:

$$\text{ConflictScore}(q) = \sum_{i=1}^{m} \alpha_i \cdot \tilde{\text{conf}}_{j_i}. \tag{4}$$

Using VideoRAG as the baseline, we compute the LLM-as-a-Judge pairwise win-rate. Queries are sorted by $\text{ConflictScore}(q)$ to plot the rolling win-rate and its 95% confidence interval, alongside the conflict score distribution to ensure sufficient high-conflict samples. In Figure 5, "Parity" denotes the tie-line with VideoRAG. The Wilson 95% confidence interval reflects local estimation uncertainty, while question counts confirm statistical stability across intervals. Across low ($\leq 0.33$), mid (0.33–0.66), and high ($> 0.66$) conflict intervals, our win-rates are 53.5% ($n = 372$), 60.7% ($n = 178$), and 58.8% ($n = 114$), respectively. The rolling curve generally ascends with increasing conflict and frequently exceeds Parity in mid-to-high conflict zones. This demonstrates that AVA consistently outperforms VideoRAG on samples with stronger cross-modal contradictions. Conversely, because the binomial win-rate is estimated locally within a fixed window, the Wilson interval widens in extreme high-conflict regions due to higher sample heterogeneity and lower density, leading to greater variance in local estimates.

### D. Ablation Studies

We further quantify the contribution of two core mechanistic designs: the agent-driven routing policy and the structured tabular indexing.

First, we replace our iterative planning-routing-feedback agent with a standard single-shot global dense retrieval step. Table III reveals a severe performance degradation: the multimodal integration win-rate plummets from 68.9% to 52.4%, VCR-style accuracy drops by over 11%, and the overall absolute score decreases from 4.09 to 3.52. This confirms that static retrieval is insufficient for long-video QA, yet adaptive tool selection and iterative feedback are important for stable evidence aggregation and multi-step reasoning over long video contexts. Importantly, while the iterative agent routing introduces computational overhead, the average inference latency per query increases by only 1.2 seconds compared to single-shot baselines, presenting a highly favorable trade-off for the substantial gains in reasoning accuracy.

TABLE III. ABLATION ON AGENT-DRIVEN RETRIEVAL.

| Metric | Full Model | w/o Agent Policy |
|---|---|---|
| Multimodal Win Rate (%) ↑ | 68.9 | 52.4 |
| VCR Accuracy (%) ↑ | 73.2 | 62.1 |
| Overall Score (1–5) ↑ | 4.09 | 3.52 |

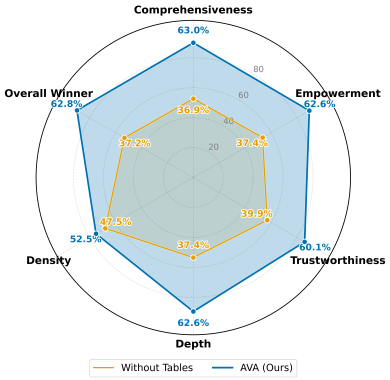



Fig. 6. Ablation results showing the impact of the tabular knowledge base.

Second, to evaluate the necessity of explicit structure, we replace the semantic tabular index with flattened, unstructured text chunks. As shown in Figure 6, structured tabular storage consistently improves answer quality, achieving an overall win-rate of 62.84% against the unstructured variant. The gains are most prominent in Comprehensiveness (63.01%) and Trustworthiness (60.14%), indicating that temporally aligned and structured evidence organization improves completeness and reliability.

## V. CONCLUSION

This paper presents AVA, an agent-driven RAG framework for long-form video understanding. By coupling a structured STC-Index with iterative dynamic retrieval, AVA effectively addresses long-range temporal dependencies and cross-modal conflicts. Experiments on the LongVideoReasoning benchmark show that AVA outperforms state-of-the-art baselines in multi-step reasoning and logical depth. Ablation studies further confirm that agent-driven routing and structured knowledge representation are both essential for maintaining temporal coherence. Future work will extend to support multi-turn interaction and real-time reranking, thereby improving scalability for large-scale video processing.

## ACKNOWLEDGMENT

The authors would like to thank the 2024 China Unicom Online Industry Intelligent Agent Key Technology Research, Development and Application Demonstration Project – Industry Intelligent Agent Module for their support.

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